Now, when order of divisors is reversed, we have:

$$\begin{array}{r|l} 13 & 2168 \\ \hline 11 & 166 - 10 \\ \hline 9 & 15 - 1 \\ \hline & 1 - 6 \end{array}$$

$\therefore$ Respective remainders are 10, 1 and 6.

**316.**
$$\begin{array}{r|l} 3 & x \\ \hline 4 & y - 2 \\ \hline 7 & z - 1 \\ \hline & 1 - 4 \end{array}$$

$z = 7 \times 1 + 4 = 11.$
$y = 4 \times z + 1 = 4 \times 11 + 1 = 45.$
$x = 3 \times y + 2 = 3 \times 45 + 2 = 137.$

When 137 is divided by 84, the remainder obtained is 53.

**317.**
$$\begin{array}{r|l} 8 & x \\ \hline 7 & y - 3 \\ \hline 3 & z - 4 \\ \hline & 31 - 2 \end{array}$$

$z = 3 \times 31 + 2 = 95.$
$y = 7 \times z + 4 = 7 \times 95 + 4 = 669.$
$x = 8 \times y + 3 = 8 \times 669 + 3 = 5355.$

$\therefore$ Required number = 5355.

**318.**
$$\begin{array}{r|l} 3 & x \\ \hline 2 & y - 1 \\ \hline & 1 - 1 \end{array}$$

$y = 2 \times 1 + 1 = 3$
$x = 3 \times y + 1 = 3 \times 3 + 1 = 10.$

Clearly, 10 when divided by 6, leaves a remainder 4.

**319.** Let the number be $N = 2x + 1$.

$N^2 = (2x + 1)^2 = 4x^2 + 1 + 4x = 4x\,(x + 1) + 1.$

Clearly, $4x\,(x + 1)$ is always divisible by 8 since one of $x$ and $(x + 1)$ is even which when multiplied by 4 is always divisible by 8.

Hence, required remainder = 1.

**320.** Sum of digits of numbers from 1 to 10 = 46.

Sum of digits of numbers from 11 to 20 = 56.

Sum of digits of numbers from 21 to 29 = 63.

Sum of digits of the given number = 46 + 56 + 63 = 165.

So, the required remainder is the remainder obtained on dividing 165 by 9, which is 3.

**321.** When $n$ is even, $(x^n - a^n)$ is divisible by $(x + a)$.

$\therefore$ $(17^{200} - 1^{200})$ is divisible by $(17 + 1)$

$\Rightarrow$ $(17^{200} - 1)$ is divisible by 18

$\Rightarrow$ On dividing $17^{200}$ by 18, we get 1 as remainder.

**322.** $2^{31} = 2 \times 2^{30} = 2 \times (2^2)^{15} = 2 \times 4^{15}$.

When $n$ is odd, $(x^n + a^n)$ is divisible by $(x + a)$.

$\therefore$ $(4^{15} + 1^{15})$ is divisible by $(4 + 1)$

$\Rightarrow$ $(4^{15} + 1)$ is divisible by 5 $\Rightarrow$ $(2^{30} + 1)$ is divisible by 5

$\Rightarrow$ On dividing $2^{30}$ by 5, we get $(5 - 1)$ i.e. 4 as remainder.

$\therefore$ Remainder obtained on dividing $2^{31}$ by 5

= Remainder obtained on dividing $(2 \times 4)$ i.e. 8 by 5 = 3.

**323.** Clearly,

(1) when $n$ is odd, $(a^n + b^n)$ is divisible by $(a + b)$. So, (1) is true.

(2) $(a^n - b^n)$ is divisible by $(a - b)$ for all values of $n$. So, (2) is true.

**324.** $(x^n - a^n)$ is divisible by $(x - a)$ for all values of $n$.

$\therefore$ $(7^{19} - 1^{19})$ is divisible by $(7 - 1)$

$\Rightarrow$ $(7^{19} - 1)$ is divisible by 6

$\Rightarrow$ On dividing $(7^{19} + 2)$ by 6, remainder obtained = 3.

**325.** $(10^{12} + 25)^2 - (10^{12} - 25)^2 = 4 \times 10^{12} \times 25$

[$\because$ $(a + b)^2 - (a - b)^2 = 4ab$]

$= 10^{12} \times 100 = 10^{12} \times 10^2 = 10^{14}$.

Hence, $n = 14$.

**326.** $(3^{25} + 3^{26} + 3^{27} + 3^{28}) = 3^{25}\,(1 + 3 + 3^2 + 3^3)$

$= 3^{25}\,(1 + 3 + 9 + 27) = 3^{25} \times 40$

$= (3 \times 10) \times (3^{24} \times 4) = 30 \times (3^{24} \times 4)$,

which is divisible by 30.

**327.** $(4^{61} + 4^{62} + 4^{63} + 4^{64}) = 4^{61}\,(1 + 4 + 4^2 + 4^3) = 4^{61} \times 85$, which is divisible by 17.

**328.** $(x^n - a^n)$ is divisible by $(x - a)$ for all values of $n$

$\therefore$ $(9^6 - 1^6)$ is divisible by $(9 - 1)$

$\Rightarrow$ $(9^6 - 1)$ is divisible by 8

$\Rightarrow$ On dividing $(9^6 + 1)$ by 8, we get 2 as remainder.

**329.** When $n$ is even, $(x^n - a^n)$ is divisible by both $(x - a)$ and $(x + a)$.

So, $(6^n - 1)$ is divisible by both $(6 - 1)$ and $(6 + 1)$

$\Rightarrow$ $(6^n - 1)$ is divisible by both 5 and 7

$\Rightarrow$ $(6^n - 1)$ is divisible by $(5 \times 7)$, i.e. 35.

[$\because$ 5 and 7 are co-primes]

**330.** $(x^n + a^n)$ is divisible by $(x + a)$ when $n$ is odd.

$\therefore$ $(12^n + 1)$ is divisible by $(12 + 1)$ i.e. 13 when $n$ is odd.

**331.** $(x^n + a^n)$ is divisible by $(x + a)$ when $n$ is odd.

$\therefore$ $(25^{25} + 1^{25})$ is divisible by $(25 + 1)$

$\Rightarrow$ $(25^{25} + 1)$ is divisible by 26

$\Rightarrow$ On dividing $25^{25}$ by 26, we get $(26 - 1) = 25$ as remainder.

**332.** $(x^n + a^n)$ is divisible by $(x + a)$ when $n$ is odd.

$\therefore$ $(67^{67} + 1^{67})$ is divisible by $(67 + 1)$

$\Rightarrow$ $(67^{67} + 1)$ is divisible by 68

$\Rightarrow$ On dividing $(67^{67} + 67)$ by 68, we get $(67 - 1) = 66$ as remainder.

**333.** $(49^{15} - 1) = (7^2)^{15} - 1 = 7^{30} - 1$.

Now, when $n$ is even, $(x^n - a^n)$ is divisible by both $(x - a)$ and $(x + a)$.

$\therefore$ $(7^{30} - 1)$ is divisible by both $(7 - 1)$ and $(7 + 1)$ i.e. by both 6 and 8. Thus, $[(49)^{15} - 1]$ is divisible by both 6 and 8.

**334.** $7^{84} = (7^3)^{28} = (343)^{28}$.

Now, $(x^n - a^n)$ is divisible by $(x - a)$ for all values of $n$.

$\therefore$ $[(343)^{28} - 1]$ is divisible by $(343 - 1)$

$\Rightarrow$ $[(343)^{28} - 1]$ is divisible by 342

$\Rightarrow$ $(7^{84} - 1)$ is divisible by 342

$\Rightarrow$ On dividing $7^{84}$ by 342, we get 1 as remainder.

**335.** $2^{60} = (2^2)^{30} = 4^{30}$.

When $n$ is even, $(x^n - a^n)$ is divisible by $(x + a)$.

$\therefore$ $(4^{30} - 1^{30})$ is divisible by $(4 + 1)$

$\Rightarrow$ $(4^{30} - 1)$ is divisible by 5 $\Rightarrow$ $(2^{60} - 1)$ is divisible by 5.

$\Rightarrow$ On dividing $2^{60}$ by 5, we get 1 as remainder.

**336.** $(2^{12} - 1) = (4096 - 1) = 4095$, which is clearly divisible by 3, 5, 7 and 13 i.e. four numbers in all.

**337.** $(x^n + a^n)$ is divisible by $(x + a)$ when $n$ is odd.

$\therefore$ $(15^{23} + 23^{23})$ is divisible by $(15 + 23)$